

Bhaumik Bhalara

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SUMMARY

Aspiring Electrical and Electronics Engineer with a strong background in Verilog, Python, and FPGA hardware design. Eager to leverage technical expertise in design verification and validation of machine learning hardware in a dynamic environment that values innovation and continuous learning.

EXPERIENCE

Design Verification Intern

Essar Power Hazira LTD, Surat, India

Jan 2022 - Mar 2022

- Gained hands-on experience with control systems utilized in boiler-actuated power generation, directly impacting the efficiency of energy production processes.
- Conducted comprehensive research on link mechanisms designed to transport waste loads from boilers, optimizing waste management systems by 15%.
- Monitored and recorded gauge and probe data while ensuring safety compliance, contributing to a 100% safety audit pass rate through participation in seminars and training sessions.

3D Printing Intern

3D Cubic, Surat, India

Aug 2021 - Oct 2021

- Completed an internship focused on understanding and operating advanced 3D printing technology, enhancing my practical skills in additive manufacturing.
- Showcased hands-on expertise by designing and fabricating a 2 cm x 2 cm x 2 cm polyamide cube using a 3D printer, demonstrating precision in design and execution.

EDUCATION

Master of Science

California State University, Sacramento, Sacramento, CA, USA

Present

GPA: 3.23/4

Bachelor of Engineering

Sarvajani College of Engineering and Technology, Gujarat, India

May 2022

GPA: 3.30/4

SKILLS

Verilog, Python, Perl, FPGA Design, Cadence, Synopsys VCS, Primetime, Quartus II IDE, MATLAB, TCL, Basic C, G-Code, PCI-E, PCI, Intel DE-10 Lite FPGA Board, Oscilloscopes, DMM, Signal Generators, Sensors, Probes, Bread Board

PROJECTS

64-bit PCI to SRAM Memory Card

Aug 2023 - Dec 2023

- Designed SRAM with detailed block and state diagrams for PCI to SRAM transactions, achieving a 20% increase in data transfer efficiency.
- Prepared timing diagrams for both 64-64 bit and 32-64-bit bus transactions, ensuring compliance with industry standards and specifications.

Vending Machine Simulation

Aug 2023 - Dec 2023

- Integrated finite state machines, including idle, selection, payment, dispensing, and reset states, to enhance user experience and operational efficiency.
- Tested and verified the design using Verilog testbenches and simulation environments, achieving a 95% success rate in functional validation.

6-Bit Segmented DAC Implementation

Jan 2024 - May 2024

- Achieved a reduction in glitch noise by adding two bits to the DAC design, resulting in improved accuracy of output signals to 12V.
- Successfully validated theoretical and simulated values through rigorous testing, ensuring design met performance specifications.

Static Timing Analysis for Microprocessor Netlist

Aug 2024 - Dec 2024

- Executed design constraints script to generate comprehensive setup and hold timing reports, identifying critical paths affecting performance.
- Identified timing violations using detailed reports and optimized slack, improving overall design performance by 30%.

4-Bit Ripple Carry Adder Verification

Aug 2024 - Dec 2024

- Implemented a Verilog module and Perl script to verify the functionality of the adder, achieving successful validation of all test cases.
- Generated detailed timing reports by applying design constraints, facilitating effective static timing analysis and ensuring design reliability.